



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: August 4, 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No: CHEM G551
Course Title: Advanced Organic Chemistry
Instructor-in-Charge: INDRESH KUMAR
Instructor: -

1. Course Description: The latest development in the area of organic chemistry will be taught in this course.

2. Scope and Objective of the Course: The course aims to cover topics in advanced areas of organic chemistry. The focus will be on the application of structure and theory to the study of organic reactions: nucleophilic aromatic substitution, organometallic reagents in organic synthesis, reaction dynamics, isotope effects, and molecular orbital theory applied to pericyclic and photochemical reactions; reactive intermediates including carbenes, carbanions, and benzyne; and asymmetric synthesis and multistep synthesis using the disconnection approach. Starting with fundamental principles and their application, the ultimate purpose of this course will be to understand the recent developments in organic chemistry.

3. Text Books: T1: "Organic Chemistry," J. Clayden, N. Greeves, S. Warren, P. Wothers, Oxford University Press, New York (2001).

4. Reference Books: R1: "Advanced Organic Chemistry: Reaction Mechanism and Structure," J. March, 4th Edition, Wiley-Interscience, New York (1992)

R2: "Organic Chemistry," R. T. Morrison, R. N. Boyd, S. K. Bhattacharjee, 7th Edition, Pearson, India (2011)

5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1-3	Some aspects of basic physical organic chemistry	class notes	Understand the basic concept of physical organic chemistry
4-5	Determining reaction mechanisms	Chap 41, T1 Chap 6, R1	The student will be able to predict the reaction mechanism of various reactions.
6-7	Enol, enolates, and nucleophilic acyl substitution; Controlling reactivity, regioselectivity, and stereoselectivity	Chap 26-28, T1	The concept of selectivity in all reactions and controlled reactivity, Chemistry of carbonyl compounds
8-9	Nucleophilic addition to the carbonyl group and Nucleophilic Aromatic Substitution, including the benzyne mechanism	Chap 6, T1, Chap 13, R1	Understand various ways of S _N Ar and nucleophilic addition to aldehydes and ketones.
10-12	Synthesis and reaction of nitrenes and carbenes as reactive intermediates	Chap 40, T1 Chap 5, R1	Basic understanding of different reaction intermediate, their generation, stabilization, and reactive patterns
13-19	Using organometallic reagents to make C-C bonds: Lithium and Magnesium reagents, other organometallic reagents (Si, Sn, B, Cu), and Transition metal-catalyzed reactions - Stille, Suzuki, Negishi, Heck, and other couplings	Chap 9, 46-48, T1	Organometallic chemistry and application of organometallic compounds in various coupling reactions



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20-24	Rearrangement reactions in organic chemistry	Chap 37, T1 Chap 18, R1	Acid or base-mediated rearrangements/ring expansion
25-30	Pericyclic reactions: The effects of orbital symmetry, Diels-Alder reactions, cycloadditions, Sigmatropic rearrangements, and Electrocyclic reactions.	Chap 35, 36, T1 Chap 20, R2	MOs correlation for various pericyclic reactions
31-35	Asymmetric synthesis: chiral reagents, chiral catalysts, chiral auxiliary, chiral pool, and resolution	Chap 45, T1	Different aspects of chiral chemistry, asymmetric synthesis from prochiral substrates
36-40	Natural products: multi-step synthesis involving protection, deprotection, and retrosynthetic analysis	Chap 51, T1 Reference articles	Designing synthetic routes for the synthesis of complex alkaloids through different disconnection approaches.

6. Evaluation Scheme:

Component	Duration	Weightage (%) (Marks)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	30 (60)	<TEST_1>	To be announced separately
Comprehensive Examination	3 h	40 (80)	<TEST_C>	-
Assignments/quizzes	-	15 (30)	Continuous	-
Seminars/Literature search	-	15 (30)	Continuous	-

7. Chamber Consultation Hour: To be announced separately.

8. Notices: Notices, if any, will be displayed on the **Nalanda & Department of Chemistry Notice Board only.**

9. Make-up Policy: Makeup would be considered only for **genuine reasons**. Make-up for continuous evaluation (assignments/quizzes) will be considered (for genuine cases) only if a student misses more than one component (assignments/quizzes).

10. Note (if any): The number of lectures shown above is notional. The students will have to rely mainly on self-study, followed by presentations and discussions. Extensive problem-solving will be an integral part of the learning process. The instructor will provide references from recent literature in specific advanced areas.

Home Assignments: Assignments will be given periodically to supplement the material discussed in class. Students will also have to deliver seminars on selected topics and collect relevant literature on them.

Prof. Indresh Kumar
Instructor-in-charge
Course No. CHEM G551